

Dive Cycle & Control Files

A Seaglider is piloted **indirectly**: the vehicle repeats a *canonical dive* — surface, dive, apogee, climb, and back to the surface — entirely on its own, and the pilot influences the *next* dive by leaving files on the basestation for the glider to collect when it calls in. There is no live terminal: everything the pilot wants to change is expressed as a handful of plain-text control files. Understanding the dive cycle and those files is the foundation of Seaglider piloting — every other topic (trim, flight model, sampling) builds on it.

Source

Paraphrased from the APL-UW IOP SGX *Documentation* (v1.0, September 2024), the UW *Seaglider Pilot's Guide*, the IOP *Basestation3 and seaglider.pub Best Practices* note, and the IOP Seaglider webinar series (iop.apl.washington.edu/iopsg). Parameter behavior varies between firmware revisions — always confirm against the [Parameter Reference Manual](#) for your firmware, and defer to the APL-UW IOP group's documentation.

The control design in one paragraph

Seaglider flight control follows two guiding principles: **hold a constant vertical velocity** and **spend as little energy as possible** doing it. The glider samples its sensors evenly in time, so constant vertical speed gives evenly spaced samples in depth. The pump that moves hydraulic oil to the external bladder is roughly **half the energy budget** of the vehicle, so the control scheme is built around avoiding unnecessary pump strokes: no bleeding is allowed on the dive (excess speed is tolerated), and pumping on the climb is allowed freely because that oil would have to be pumped at the surface anyway.

The desired vertical velocity is never set directly. It falls out of two parameters:

```
w_desired = (2 × $D_TGT × 100 cm/m) / ($T_DIVE × 60 s/min) [cm/s]
```